Last Time

$\mathcal{H} := L^2(X_0, d\lambda_0, \mathbb{C}^4) = \{ f : X_0 \to \mathbb{C}^4 : f = (f_0, f_1, f_2, f_3)$ with $\int_{X_0} |f_i|^2 d\lambda_0 < \infty \}$.

$(\xi, \psi) := -\int d\lambda_0(p) \, (\xi(p), \psi(p))$ for $\xi, \psi \in \mathcal{H}$.

↳ Inner Pr. is <u>not</u> positive-definite.

$\mathcal{H}' := \{ \xi \in \mathcal{H} : (p \cdot \xi(p)) = 0$ for λ_0-a.e. $p \}$.

⇒ Inner Pr. is positive-semidefinite.

$\mathcal{H}_{null} := \{ \xi \in \mathcal{H}' : (\xi, \xi) = 0 \}$.

$\mathcal{H}_{phys} := \mathcal{H}' / \mathcal{H}_{null}$: Hilbert sp. under Minkowski.

posi-def.

Hilbert sp. for a single photon.

$(a, A) \in \mathcal{P} = \mathbb{R}^{1,3} \rtimes SO^{\uparrow}(1,3)$.

For $\psi \in \mathcal{H}$, define

$$(\mathcal{U}(a,A)\underbrace{\psi}_{\text{sys. state}})(p) = e^{i(a,p)} A \psi(A^{-1}p)$$: representation of $\mathcal{P}$ in $\mathcal{H}$.

sys. state with new coord.

$(a.A) \in \mathcal{P} = \mathbb{R}^{1,3} \rtimes SO^{\uparrow}(1,3).$

For $\psi \in \mathcal{H}$, define

$$\underbrace{(U(a.A)\psi)}_{\text{system state}}(p) = e^{i(a,p)} A\,\psi(A^{-1}p).$$

system state

after changing the system state.

coordi. sys. by the action of (a,A).

$\to$ representation of $\mathcal{P}$ in $\mathcal{H}$.

<u>Fact</u> $U(a.A) : \mathcal{H}_{phys} \to \mathcal{H}_{phys}.$

λ_0: Lorentz-invariant.

<pf> 1. $\mathcal{H}' \to \mathcal{H}'$.

Recall that $\mathcal{H}' := \{ \psi : (p.\psi(p)) = 0, \text{ for } \lambda_0 \text{ a.e. } p \}$.

$$\int d\lambda_0(p)(p.\,U(a.A)\psi(p)) = \int d\lambda_0(p)(p.\,e^{i(a,p)}A\psi(A^{-1}p))$$

$$= \int d\lambda_0(p)\,e^{i(a,p)}\underbrace{(A^{-1}p.\,\psi(A^{-1}p))}_{0 \text{ for } \lambda_0 \text{ a.e.p.}} = 0$$

$\Rightarrow (p.\,U(a.A)\psi(p)) = 0$, for λ_0 a.e. p.

$\therefore U(a.A)\psi \in \mathcal{H}'.$

2. $\mathcal{H}_{phys} \to \mathcal{H}_{phys}$. (i.e., $\mathcal{H}_{null} \to \mathcal{H}_{null}$)

$(U(a.A)\psi,\, U(a.A)\xi)$

$$= -\int d\lambda_0(p)\,(e^{i(a.p)}A\psi(A^{-1}p),\, e^{i(a,p)}A\xi(A^{-1}p))$$

$$= -\int d\lambda_0(p)\,(A\psi(A^{-1}p),\, A\xi(A^{-1}p))$$

$$= -\int d\lambda_0(p)\,(\psi(A^{-1}p),\, \xi(A^{-1}(p))$$

$$= -\int d\lambda_0(p)\,(\psi(p),\, \xi(p)) = (\psi, \xi). \quad \square$$

Inner product is Lorentz-invariant.

λ_0 is Lorentz-invariant.

The electromagnetic four-potential

$\mathcal{H} = L^2(X_0, d\lambda_0, \mathbb{C}^4)$ with Euclidean inner product.

$$(\xi, \psi)_E = \int d\lambda_0(p) \sum_{\mu=0}^{3} \xi^\mu(p)^* \psi^\mu(p).$$

Euclidean

$\Rightarrow \mathcal{H}$: Hilbert space under this inner product.

$\mathcal{B} :=$ Fock space of $\mathcal{H}$.

$=$ set of seqs $(\psi_0, \psi_1, \cdots)$, $\psi_n \in \mathcal{H}^{\otimes n}_{sym.}$ with $\sum_{n=0}^{\infty} \|\psi_n\|^2 < \infty$, $\forall n$.

$\mathcal{H}$: basis $\{e_i\}$

basis : $\{e_{i_1} \otimes \cdots \otimes e_{i_n}\}$

$= e_{\sigma(i_1)} \otimes \cdots \otimes e_{\sigma(i_n)}$

Lec7

Define $B(\xi) = A(-\eta\xi)$, $B^+(\xi) = A^+(\xi)$

where $\eta = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$.

$\rightarrow B, B^+$: operator-valued distributions on $\mathcal{H}$.

$$[B(\xi), B(\psi)] = [B^+(\xi), B^+(\psi)] = 0$$

$$[B(\xi), B^+(\psi)] = [A(-\eta\xi), A^+(\psi)] = (-\eta\xi, \psi)_E \mathbb{1}$$
$$= (\xi, \psi)\mathbb{1}.$$

For $f \in \mathcal{S}(\mathbb{R}^4)$, $\mu = 0, 1, 2, 3$, define

$\mathcal{S}(\mathbb{R}^4)$: space of Schwartz ft on $\mathbb{R}^4$.

$\hat{f}$: Fourier transform

$$A_\mu(f) = B(\hat{f} e_\mu) + B^+(\hat{f} e_\mu). \in \mathcal{B}_0.$$

e_μ : μth standard basis vector of $\mathbb{R}^{1,3}$.

$A = (A_0, A_1, A_2, A_3)$: quantized electromagnetic four-potential.

(※) $A(f) = \sum_{k=1}^{\infty} \alpha_k^* a_k$ $\quad$ $A^+(f) = \sum_{k=1}^{\infty} \alpha_k a_k^+$

$f = \sum_{k=1}^{\infty} \alpha_k e_k$

$a_k^+ |n_1, n_2, \cdots\rangle = \sqrt{n_k+1}\, |n_1, n_2, \cdots, n_{k-1}, n_k+1, n_{k+1}, \cdots\rangle$

where $a_k |n_1, n_2, \cdots\rangle = \begin{cases} \sqrt{n_k}\, |n_1, n_2, \cdots, n_{k-1}, n_k - 1, n_{k+1}, \cdots\rangle, & n_k \geq 1. \\ 0, & n_k = 0. \end{cases}$

Hilbert space $\mathcal{H}$, $\mathcal{B} := \{ (\psi_0, \psi_1, \cdots) : \psi_n \in \mathcal{H}^{\otimes n}_{sym}$ with $\sum_{n=0}^{\infty} \|\psi_n\|^2 < \infty \}$.

$$(\psi, \phi) = \sum_{n=0}^{\infty} (\psi_n, \phi_n).$$

$\Rightarrow \mathcal{B}$: Hilbert space. $\rightarrow$ Bosonic Fock sp of $\mathcal{H}$.

24. The Electro Magnetic Field.

$\mathcal{B}$: bosonic Fock space of $\mathcal{H}$.
$\mathcal{B}'$: " of $\mathcal{H}'$.
$\Rightarrow$ $\mathcal{B}'$: closed subspace of $\mathcal{B}$.

Define
$$(\xi_1 \otimes \cdots \otimes \xi_n, \psi_1 \otimes \cdots \otimes \psi_n) := \prod_{i=1}^{n} (\xi_i, \psi_i).$$

$\mathcal{B}_{null} := \{ \psi \in \mathcal{B}' : \|\psi\| = 0 \}$.
$\mathcal{B}_{phys} := \mathcal{B}' / \mathcal{B}_{null}$ $\underset{iso}{\cong}$ Bosonic Fock sp of $\mathcal{H}_{phys}$.

For $f \in \mathcal{S}(\mathbb{R}^4)$, $A_\mu(f)$ acts on $\mathcal{B}$, $\mu = 0, 1, 2, 3$.
But $A_\mu(f)$ not acts on $\mathcal{B}_{phys}$.
Because $A_\mu(f)$ not acts on $\mathcal{B}_{null}$.
Some linear combination of A_μ's are well-defined on $\mathcal{B}_{phys}$.
ex1) $F_{\mu\nu} := \partial_\mu A_\nu - \partial_\nu A_\mu$. $\left(\partial_\mu = \dfrac{\partial}{\partial x^\mu} \right)$
$\Rightarrow$ For any $f \in \mathcal{S}(\mathbb{R}^4)$, $F_{\mu\nu}(f)$ is well defined on $\mathcal{B}_{phys}$.

$F = (F_{\mu\nu})_{0 \le \mu, \nu \le 3}$: "electromagnetic field."

$$\begin{bmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & -B_z & B_y \\ -E_y & B_z & 0 & -B_x \\ -E_z & -B_y & B_x & 0 \end{bmatrix}$$

Electro field $E = (E_x, E_y, E_z)$
Magnetic field $B = (B_x, B_y, B_z)$

ex2) $f_0 . f_1 . f_2 . f_3 \in \mathcal{S}(\mathbb{R}^4)$ s.t

$$\partial_0 f_0 - \partial_1 f_1 - \partial_2 f_2 - \partial_3 f_3 = 0.$$

$\Rightarrow$ $A_0(f_0) - A_1(f_1) - A_2(f_2) - A_3(f_3)$ acts on $\mathcal{B}_{phys}$.

› Classical (Maxwell) theory of electromagnetism.

In Maxwell's formulation, we have two vector fields.

$\mathbb{E} : \mathbb{R}^{1,3} \longrightarrow \mathbb{R}^3$ "electric field"

$\mathbb{B} : \mathbb{R}^{1,3} \longrightarrow \mathbb{R}^3$ "magnetic field"

<u>Maxwell eqs.</u> (진공상)

divergence

(1) $\nabla \cdot \mathbb{E} = 4\pi\rho$ charge density.

(2) $\nabla \cdot \mathbb{B} = 0$

curl

(3) $\nabla \times \mathbb{E} = -\frac{\partial \mathbb{B}}{\partial t}$

(4) $\nabla \times \mathbb{B} = 4\pi \mathbb{J} + \frac{\partial \mathbb{E}}{\partial t}$

current density

$\rho, \mathbb{J}$ determine the evolution of electric and magnetic fields.

Assume no electron. i.e., $\rho \equiv 0$, $\mathbb{J} = 0$.

$\rightarrow$ Maxwell's eqs in the vacuum.

classical electromagnetic four-potential.

Take any smooth vector field

$A : \mathbb{R}^{1,3} \longrightarrow \mathbb{R}^4$. $A = (A_0 , A_1 , A_2 , A_3)$

Define the "classical electromagnetic field" $F = (F_{\mu\nu})_{0 \le \mu,\nu \le 3}$

as $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

Define two v.f. E & B as $E = (E_x , E_y , E_z)$

$B = (B_x , B_y , B_z)$

$$F = \begin{bmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & -B_z & B_y \\ -E_y & B_z & 0 & -B_x \\ -E_z & -B_y & B_x & 0 \end{bmatrix}$$

By the def. of F,
We have an identity, called "Bianchi identity".

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0, \quad \forall \mu, \lambda, \nu.$$

And the v.f. B & E satisfy (2), (3) in Maxwell eq.

- Gauss's law for magnetism. (2)

$$\begin{aligned}\nabla \cdot B &= \partial_x B_x + \partial_y B_y + \partial_z B_z \\ &= -\partial_x F_{23} + \partial_y F_{13} - \partial_z F_{12} \\ &= \partial_x F_{32} + \partial_y F_{13} + \partial_z F_{21} \\ &= \partial_1 F_{32} + \partial_2 F_{13} + \partial_3 F_{21} = 0.\end{aligned}$$

$x=1, y=2, z=3$

- Maxwell-Faraday equation. (3)

$$\nabla \times E = \begin{vmatrix} i & j & k \\ \partial_x & \partial_y & \partial_z \\ E_x & E_y & E_z \end{vmatrix}$$

$$= (\partial_y E_z - \partial_z E_y, -\partial_x E_z + \partial_z E_x, \partial_x E_y - \partial_y E_x)$$

$$\partial_y E_z - \partial_z E_y = \partial_y F_{tz} - \partial_z F_{ty} = \partial_y F_{tz} + \partial_z F_{yt} = \partial_t F_{yz}$$
$$= \partial_t F_{23} = \partial_t(-B_x)$$

$$\therefore = -(\partial_t B_x, \partial_t B_y, \partial_t B_z).$$

Adding the condition, so called "equation of evolution"

$$\partial_0 F_{0\nu} - \partial_1 F_{1\nu} - \partial_2 F_{2\nu} - \partial_3 F_{3\nu} = 0, \quad \forall \nu.$$

we can see B & E also satisfy (1), (4) in Maxwell equations in the vacuum.

- Gauss's Law.

$$\nabla\cdot E = \partial_x E_x + \partial_y E_y + \partial_z E_z$$
$$= \partial_x F_{tx} + \partial_y F_{ty} + \partial_z F_{tz}$$
$$= -\partial_x F_{xt} - \partial_y F_{yt} - \partial_z F_{zt} = -\partial_t F_{tt} = 0$$

- Ampère's Circuital Law.

$$\nabla\times B = \begin{vmatrix} i & j & k \\ \partial_x & \partial_y & \partial_z \\ B_x & B_y & B_z \end{vmatrix} = (\partial_y B_z - \partial_z B_y, -\partial_x B_z + \partial_z B_x, \partial_x B_y - \partial_y B_x)$$

$$\partial_y B_z - \partial_z B_y = \partial_y(-F_{xy}) - \partial_z F_{xz} = \partial_y F_{yx} + \partial_z F_{zx}$$
$$= \partial_t F_{tx} - \partial_x F_{xx}$$
$$= \partial_t E_x$$

$$= (\partial_t E_x, \partial_t E_y, \partial_t E_z).$$

<u>Fact</u> Any solution of Maxwell's equations in the vacuum is representable in the above manner.
But it's not unique.
The procedure for choosing a unique representative is called "guage fixing".
One way to do this is to impose "Lorenz gauge condition",

$$\partial_0 A_0 - \partial_1 A_1 - \partial_2 A_2 - \partial_3 A_3 = 0.$$

$\therefore$ $A = (A_\mu)_{0 \le \mu \le 3}$ defined by using Gupta–Bleuler.
satisfies the vacuum Maxwell equations
and the Lorenz gauge condition.

Einstein Notation (Summation Convention).

$$y = \sum_{i=1}^{3} c_i x^i \quad \Rightarrow \quad y = c_i x^i.$$

For (a_0, a_1, a_2, a_3),

$$a^0 := a_0, \qquad a^\mu := -a_\mu, \quad \mu = 1, 2, 3.$$

$$\partial^\mu A_\mu := \sum_{\mu=0}^{3} \partial^\mu A_\mu = \partial_0 A_0 - \partial_1 A_1 - \partial_2 A_2 - \partial_3 A_3.$$

ex) Lorenz gauge condition : $\partial^\mu A_\mu = 0$.
Minkowski inner product : $x^\mu y_\mu$.

가우스 법칙 $\nabla \cdot D = \rho$ Gauss's law

가우스 자기 법칙 $\nabla \cdot B = 0$ Gauss's law for magnetism

패러데이 전자기 유도 법칙 $\nabla \times E = -\frac{\partial B}{\partial t}$ Maxwell-Faraday eq.

앙페르-맥스웰 회로 법칙. $\nabla \times H = J + \frac{\partial D}{\partial t}$. Ampère's circuital law.

E: 전기장, H: 자계강도 D: 전기변위장 B: 자기장.

electric field magnetic field.

ρ : 자유전하밀도 J: 자유전류밀도

charge density electric current density.

진공상의 전자기장.

$$\nabla \cdot \underset{\text{전기장}}{E} = \frac{\rho_v}{\varepsilon_0} \qquad \nabla \cdot \underset{\text{자기장}}{B} = 0$$

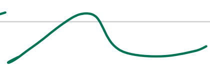
volume charge density

$$\nabla \times E = -\frac{\partial B}{\partial t} \qquad \nabla \times B = \mu_0 \left(J_v + \varepsilon_0 \frac{\partial E}{\partial t} \right).$$

volume current density.

C

$$S : \sup_{\alpha, \beta, x} |x^\beta \partial_x^\alpha f| < C$$

e^{-x^2} 

$\mathcal{D} \to S \qquad S \subseteq \mathcal{H}.$

$C_c^\infty \subseteq S \subseteq L^2.$

dense